Escola Superior d'Enginyeries Industrial, Aeroespacial i Audiovisual de Terrassa

Vehicles Aeroespacials - 220306

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1.1 Theory: Telescope Segment

Structural analysis is concerned with predicting the performance of structures by means of,

- Evaluation of **deformations** and **stresses** arising within a solid object under the action of applied **loads**. For this we will be using static structural analysis: $Ku = f$
- Studying the **dynamic behaviour** of the solids. For this we will be using dynamic structural analysis: $M\ddot{u}+D\dot{u}+Ku = f$, instead of $Ku = f$. However, it is difficult to realistically model the damping and therefore is nto taken into account when possible.
- Studying the **stability** of the structures, $K(u)\,u = f$

This is the basis of structural analysis. We will approach the problem academically and with the tools of the industry.

1.1.1 Elastic problem - Governing equations

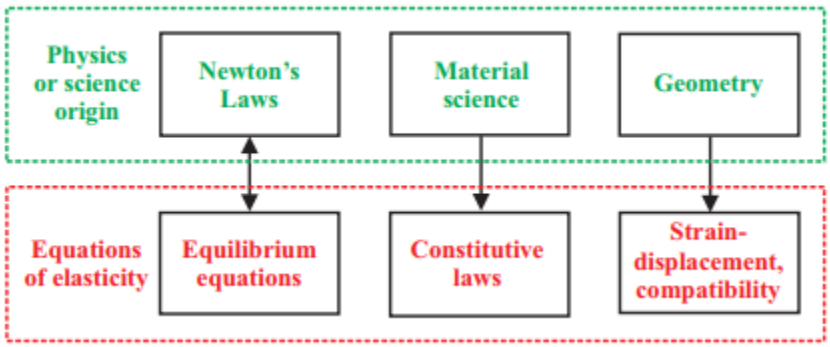

Figure 1: Elasticity equations

- **Equilibrium equation**. The equilibrium equations express the equilibrium conditions for a **differential element** of the body in terms of the **stress field**. These equilibrium conditions are a direct consequence **Newton's laws** applied to a **differential element** of the deformable body.
- **Constitutive law**. Describe the **behavior of materials** under **load**. More specifically, they take the form of relationships linking the **stress** and **strain** components at a point. Constitutive laws are rooted in **material science** and express an approximation to the observed behavior of actual materials.
  - Linear materials, plasticity, anysotropic (composites), etc.
- **Kinematic equation**. Describe the deformation of the body without reference to the forces that create the deformation. The strain components are defined based on a purely kinematic description of the deformed and undeformed configurations of the solid.

1.1.2 Elastic problem - Boundary conditions

- **Dirichlet** [1] boundary conditions, given displacements, unknown reaction force $u\mid_{\Gamma_D} = \vec{u_o}$.
- **Neumann**[2] boundary conditions, given reaction force, unknown displacement $\sigma \cdot n \mid_{\Gamma_N} = \vec{t}$. Notice that non applied load is still a known applied force, and it is not a known displacement.

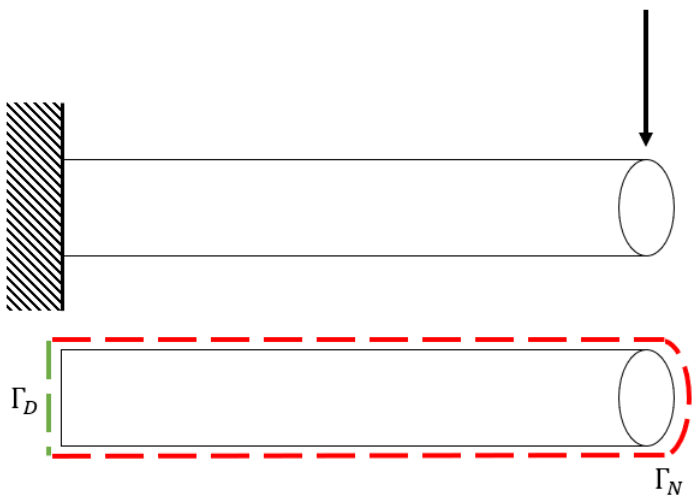

Figure 2: Boundary conditions example

1.1.3 Finite element method

We aim to convert a sistem of differential equations, which we cannot solve in a computer, into a system of algebraic equations.

$$\begin{cases} \nabla \cdot \boldsymbol{\sigma} + b = 0 \\ \sigma = f(\varepsilon) \\ \varepsilon = f(u) \end{cases} \rightarrow \boldsymbol{K}u = F \tag{1}$$

We discretize our domain into smaller parts, elements. We solve the differential equations for each element assuming that displacement follow a polynomial function. With this, the convert our differential equations with an algebraic equation which only depends on the displacements and forces in the boundaries of the element. Therefore, the solutions in each element are approximate. The smaller the element, the most accurate the approximation.

1.1.4 Sources of non linearity

There are three sources of nonlinearity in structural mechanics simulations,

- **Material nonlinearity**, plasticity, viscoelastic materials such as rubbers.

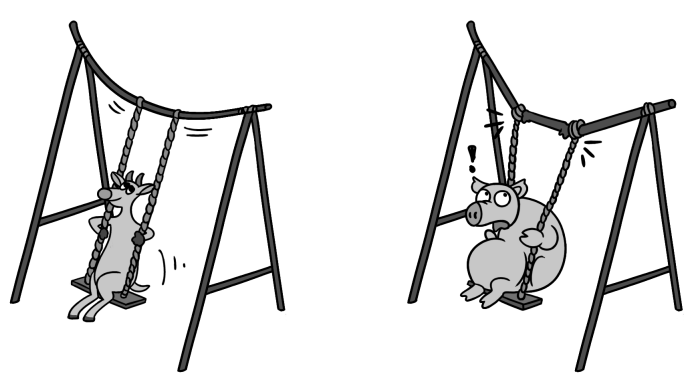

Figure 3: Beam within the elastic range (left), beam past the elastic range (right)

[1] https://en.wikipedia.org/wiki/Peter_Gustav_Lejeune_Dirichlet
[2] https://en.wikipedia.org/wiki/Carl_Neumann

- **Boundary nonlinearity**, contacts.

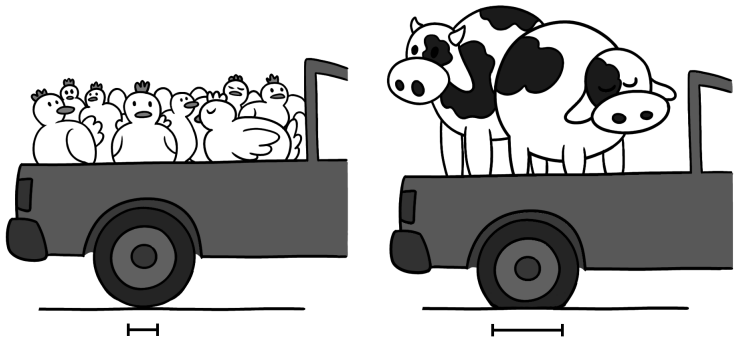

Figure 4: Contact area of a tyre depending on the load

- **Geometric nonlinearity**. The equilibrium is no longer in the undeformed state. Buckling analysis.

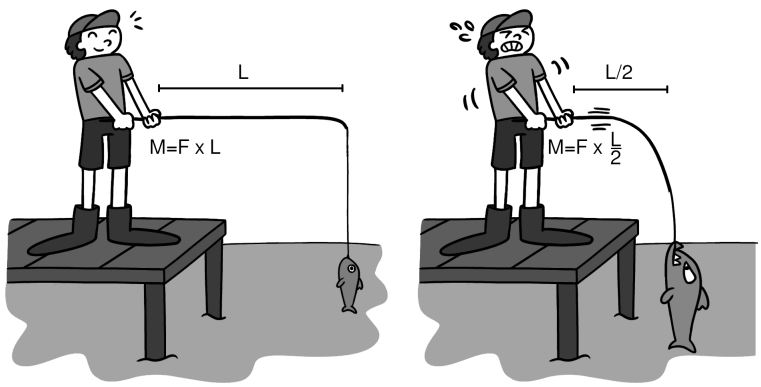

Figure 5: Geometric non linearity example

For most situations, it is not possible to develop analytical solutions of these equations. Consequently, structural analysis is concerned with the analysis of structural components, such as bars, beams, plates, or shells

1.1.5 Linear static solution

We begin with an assembled stiffness matrix $\boldsymbol{K}$. This matrix represents the relationship between the force and displacements in all the structure nodes. There are different ways to organize the stiffness matrix, but let's begin with a simple way.

- For a general structure in 3D, each node can move in 6 degrees of freedom, 3 translations and 3 rotations. Therefore, there are 6 degrees of freedom per node, $n_{dofpn} = 6$.
- Therefore, the total degrees of freedom of the structure will be the number of nodes times 6.
- The size of $\boldsymbol{K}$ will be $[n_{nodes} \cdot n_{dofpn}, n_{nodes} \cdot n_{dofpn}]$
- The size of u and F will be$[n_{nodes} \cdot n_{dofpn}, 1]$
- The displacements will be arranged in the following criteria. We will use the notation $u_{node}^{component}$.

$$u = \begin{bmatrix} u_1^1 & u_1^2 & u_1^3 & u_1^4 & u_1^5 & u_1^6 & u_2^1 & u_1^2 & ... & u_{nnodes}^5 & u_{nnodes}^6 \end{bmatrix}^T \tag{2}$$

- The same criteria is applied for the forces and the stiffness matrix.

Each degree of freedom can be classified as a Neumann or Dirichlet exclusively. Neumann degrees of freedom are those with known external forces, and Dirichelt dof are those with enforced displacements. In our code we can define some user friendly way of defining the boundary conditions. To take advanatge of MATLAB array indexing capabilities, it is recommendable to define a vector of Neumann degrees of freedom $in_N$and a vector of Dirichlet degrees of freedom in_D.

We can then partition our system of equations according to the type of degrees of freedom.

$$\boldsymbol{K}u = F \tag{3}$$

$$\begin{bmatrix} \boldsymbol{K}_{DD} & \boldsymbol{K}_{DN} \\ \boldsymbol{K}_{ND} & \boldsymbol{K}_{NN} \end{bmatrix} \begin{bmatrix} u_D \\ u_N \end{bmatrix} \begin{bmatrix} F_D \\ F_N \end{bmatrix} \quad \text{unknown} \tag{4}$$

According to this partition,

- $\boldsymbol{K}$, u_D(the displacements of the constrained nodes) and F_N(the external forces applied to the unconstrained nodes) are known.
- u_N(the displacements of the unconstrained dof) and F_D(the reactions of the constrained dof) are unknown.

And we can solve the unknowns easily,

$$\boldsymbol{K}_{NN}u_N = [F_N - \boldsymbol{K}_{ND}u_D] \tag{5}$$

$$u_N = \boldsymbol{K}_{NN}^{-1}[F_N - \boldsymbol{K}_{ND}u_D] \tag{6}$$

$$F_D = \boldsymbol{K}_{DD}u_D + \boldsymbol{K}_{DN}u_N \tag{7}$$

Although we use the notation $\boldsymbol{K}_{NN}^{-1}$ in Equation6, tipically the inverse of the stiffness matrix is never computed for numerical reasons and computational cost. We use numerical algorithms to solve the system of equations. MATLAB backslahsh[3] operator is a wrapper for many algorithms to solve linear system of equations. The backslash function chooses the algorithm dependind on the characteristics of the input matrix.

There might be applied forces at the constrained nodes. This forces do not participate in the system of equations but shall be taken into account in order to have a consistent set of reactions that are in static equilibrium with the applied forces. Be $F_{ext}$the complete set of applied forces to the structure regardless of the type of boundary conditions.

- $F_N$will consist on the subset of F_{ext} that corresponds to the Neumann degrees of freedom (F_{ext}^N).
- F_{ext}^Dwill consist on the subset of F_{ext} that corresponds to the Dirichlet degrees of freedom.

The reaction forces r_f can be therefore computed after the linear systems of equations is solved,

$$r_f = F_D - F_{ext}^D \tag{8}$$

[3] https://es.mathworks.com/help/matlab/ref/mldivide.html